

Article

A Trust Score-Based Access Control Model for Zero Trust Architecture: Design, Sensitivity Analysis, and Real-World Performance Evaluation

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Abstract

As digital infrastructures become increasingly dynamic and complex, traditional static access control mechanisms are no longer sufficient to counter advanced and persistent cyber threats. In response, Zero Trust Architecture (ZTA) emphasizes continuous verification and context-aware access decisions. To realize these principles in practice, this study introduces a Trust Score (TS)-based access control model as a systematic alternative to legacy, rule-driven approaches that lack adaptability in real-time environments. The proposed TS model quantifies the trustworthiness of users or devices based on four core factors—User Behavior (B), Network Environment (N), Device Status (D), and Threat History (T)—each derived from measurable operational attributes. These factors were carefully structured to reflect real-world Zero Trust environments, and a total of 20 detailed sub-metrics were developed to support their evaluation. This design enables accurate and granular trust assessment using live operational data, allowing for fine-tuned access control decisions aligned with Zero Trust principles. A comprehensive sensitivity analysis was conducted to evaluate the relative impact of each factor under different weight configurations and operational conditions. The results revealed that B and N are most influential in real-time evaluation scenarios, while B and T play a decisive role in triggering adaptive policy responses. This analysis provides a practical basis for designing and optimizing context-aware access control strategies. Empirical evaluations using the UNSW-NB15 dataset confirmed the TS model's computational efficiency and scalability. Compared to legacy access control approaches, the TS model achieved significantly lower latency and higher throughput with minimal memory usage, validating its suitability for deployment in real-time, resource-constrained Zero Trust environments.

Keywords: trust score; zero trust architecture; adaptive access control; threat detection; security risk assessment; trust evaluation



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1. Introduction

The traditional perimeter-based security model operates by establishing clear boundaries between internal and external networks and trusting the internal network. This model is based on the traditional view of security, where internal users are trusted and only external access is considered a potential threat. However, with cloud computing, increased use of mobile devices, the proliferation of remote work environments, and the sophistication of cyber threats, the limitations of this perimeter-based security model are becoming clear [1,2].

Insider threats, account takeover attacks, ransomware, and more exploit vulnerabilities in the traditional model of trusting internal networks, accelerating the shift to a zero trust security model.

Zero Trust Architecture (ZTA) is based on the principle of “Never Trust, Always Verify” and performs thorough verification of all access requests before granting trust [3]. It is an approach that removes the distinction between inside and outside the network and continuously evaluates the trustworthiness of users and devices to enhance security. To effectively implement ZTA, real-time threat assessment and dynamic authentication enforcement are essential [4].

In a Zero Trust environment, access control is a key element of security, performing the critical function of evaluating and determining who, where, and how to access. Traditional static access control approaches such as role-based access control (RBAC), attribute-based access control (ABAC), and risk-based authentication (RBA) are limited by their inability to reflect rapidly changing security threats and user behavior in real time [5,6]. To overcome these limitations, this research proposes dynamic authentication enforcement through trust score evaluation.

Trust Score comprehensively evaluates various factors such as user and device status, network environment, and past threat history to dynamically adjust access rights in real time. This enables continuous authentication by continuously monitoring user activity and changes in the environment, rather than simply authenticating at a point in time. By implementing flexible security policies that enforce or restrict access privileges based on context, organizations can quickly respond to changing security threats [7,8].

In this paper, we derive the main factors for calculating the Trust Score and propose a method for calculating the Trust Score based on them. In particular, we propose a mathematical model for calculating the Trust Score by systematically analyzing factors such as user behavior, network environment, device status, and past threat history. We also analyze the impact on the Trust Score due to changes in weight values through simulation and theoretical analysis for various scenarios, and propose customized weight setting information considering security factors specific to each industry.

The results of this study will provide practical guidance for organizations that want to adopt Zero Trust Architecture. In particular, it is expected to contribute to improving security through real-time threat detection and dynamic authentication enforcement, while optimizing operational efficiency and user experience.

Section 2 analyzes the differences between the existing access control method and the trust score-based access control method in the Zero Trust Architecture environment, and introduces the research background and related research that support the need for this research. Then, in Section 3, we derive the details of the major factors for calculating the Trust Score value, and introduce the mathematical modeling of the Trust Score model and the calculation examples for each major scenario. In Section 4, we verify the effectiveness of the model through simulation and theoretical analysis, and in Section 5, we assess the computational performance and scalability of the proposed Trust Score model and compare it with legacy models, demonstrating its suitability for real-time deployment. Finally, Section 6 concludes with conclusions and future research directions to discuss the potential for further development of trust score-based access control and its applicability to various industries.

2. Background and Related Work

2.1. Traditional Access Control

2.1.1. Role-Based Access Control (RBAC)

Role-Based Access Control (RBAC) is a method of granting access based on a user's role. Users are managed to access specific resources based on their defined roles within the organization. While RBAC provides simplicity of management and policy consistency, it is inflexible to dynamic security needs due to static permission settings [9].

Singh et al. [10] propose a framework to leverage RBAC in Internet of Things (IoT) networks to improve the security and efficiency of data processing. Since it grants authorization based on predefined roles, it lacks dynamic response by failing to reflect the real-time state changes of IoT devices or networks.

Butt et al. [11] present an RBAC mechanism that monitors user behavior in an electronic health (*e-health*) cloud environment and assigns roles based on trust values. Although this study applies a trust-based RBAC model, it remains limited in reflecting fine-grained changes in user states in real time due to its role-centric nature, and lacks continuous validation.

Marquis [12] investigates how effective RBAC systems are in mitigating insider threats in different organizational environments. The study proposes dynamic extensions to RBAC, but still fails to overcome the static nature of authorization management, which makes real-time threat response difficult.

Mpamugo et al. [13] explore the integration of RBAC with the OAuth 2.0 protocol in mobile applications to manage network access and enhance security. Mobile networks are characterized by rapid environmental changes, which has not been reflected in the dynamic changes in the mobile environment and is vulnerable to anomalous login attempts or location-based anomaly detection.

Carvalho Junior et al. [14] highlight the implementation characteristics, concerns, and limitations of RBAC in health information systems; specifically, they address security trends and limitations related to emergency access, authorization delegation, and cross-domain access control. In healthcare environments, emergency situations require rapid authorization changes, but RBAC's static authorization structure does not support this quickly enough.

2.1.2. Attribute-Based Access Control (ABAC)

Attribute-Based Access Control (ABAC) is an approach to determining access rights based on attributes of users, resources, and environments. ABAC allows for policy flexibility and fine-grained access control, but complex policy management is considered a drawback [15].

Servos et al. [16] provided a comprehensive review of the concept and recent research trends in ABAC, highlighting issues such as delegation, management, auditing, and scalability. However, ABAC relies on predefined attribute values, which makes it difficult to reflect changes in attribute values immediately and limits real-time threat detection and response.

Singh et al. [17] reviewed ABAC models in cloud computing environments and attempted to overcome the limitations of existing models by proposing attribute and user revocation mechanisms. Although they proposed user and attribute revocation mechanisms, the problem remains that attribute changes are not immediately reflected in security policies.

Fernandez et al. [18] proposed a category-based model to streamline attribute management in ABAC systems and simplify the definition and management of access control policies. While category-based models increase management efficiency through attribute

grouping, they have limitations in reflecting continuous changes in the environment in real time.

Panende et al. [19] evaluated the security and efficiency of access control models by comparing *ABAC* and *RBAC* in Digital Evidence Storage (*DES*). While attribute-based enables fine-grained control, conflicts between attributes complicate management, and complexity exists in updating and managing policies.

Xu et al. [20] proposed an Attribute-Based Access Control (*ABAC*) mechanism that considers both privacy and efficiency. *EPABAC* focuses on granting access rights based on attributes, but it has limited ability to reflect real-time changing user behavior or network environment.

2.1.3. Risk-Based Authentication (*RBA*)

Risk-Based Authentication (*RBA*) dynamically adjusts authentication strength by assessing the risk level of the user and environment in real time. *RBA* balances user convenience and security, but risk assessment accuracy is a key challenge [21].

Makowski et al. [22] re-evaluated the *RBA* systems of major online services through black-box testing, revealing differences based on account creation and rare activations, highlighting their complexity and limitations, and suggesting future research directions.

Büttner et al. [23] analyzed how *RBA* mechanisms are applied during account recovery in a major online service and presented a maturity model for *RBA*. Security vulnerabilities can exist in the account recovery process.

Unsel et al. [24] explored how to incorporate risk-based access control into the authentication process in OpenStack. Their approach assesses risk at the point of login and requires additional authentication, but does not perform continuous authentication after login and has limited capabilities for real-time detection and immediate access blocking.

Wiefeling et al. [25,26] evaluated the security and usability of *RBA* systems in large-scale online services and proposed improvements through machine learning-based optimization. They operate based on predefined risk assessment criteria but lack the ability to adjust weights or thresholds in real time based on changing circumstances, which is a challenge.

2.2. Zero Trust Security Models

2.2.1. Zero Trust Architecture

Key elements of Zero Trust security include strict access controls, least privilege access, micro-segmentation, and continuous monitoring and validation. Collectively, these elements significantly improve network security by minimizing the attack surface and preventing lateral movement within the network [3].

NIST 800-207 [27] defines Zero Trust Architecture as a framework consisting of the elements necessary for security policy decisions and enforcement. Figure 1 below illustrates the logical components of a Zero Trust Architecture and their interactions. This conceptual diagram illustrates the interaction of the Policy Decision Point (*PDP*) and Policy Enforcement Point (*PEP*), which are the core elements of the Zero Trust Architecture.

The Policy Decision Point (*PDP*) functions as a key component for evaluating access requests and making security decisions based on predefined policies and real-time contextual information. *PDPs* collect data from a variety of sources, including user identity, device state, network environment, and behavioral patterns. This information is analyzed to determine whether an access request should be granted, denied, or require additional authentication, and the *PDP* forwards the results to a Policy Enforcement Point (*PEP*) for enforcement. This continuous assessment enables you to maintain a flexible and adaptive security posture against internal and external threats.

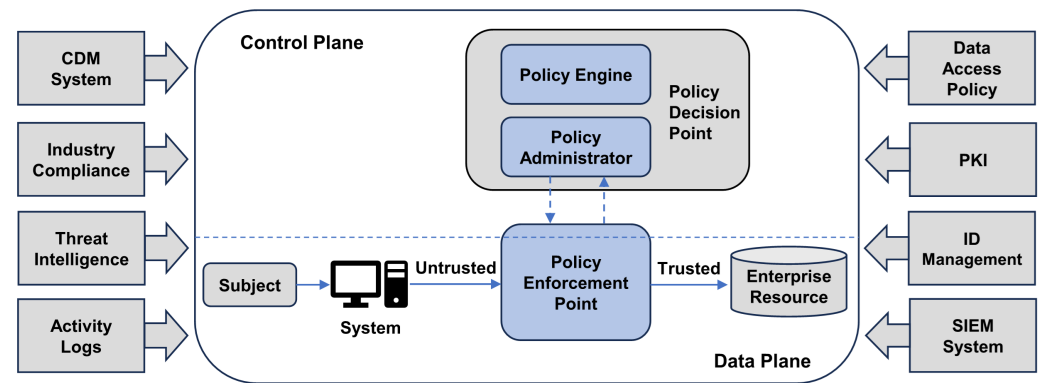


Figure 1. Zero Trust Architecture logical concepts as defined by NIST 800-207.

Policy Enforcement Points (*PEPs*) are responsible for enforcing access control decisions made by the *PDP*. *PEPs* are deployed at the network perimeter or within specific applications, intercepting access requests and forwarding them to the *PDP* for evaluation. Upon receiving the *PDP*'s decision, the *PEP* takes appropriate actions, such as allowing, denying, or requiring additional authentication for the request. Additionally, *PEPs* continuously monitor active sessions and can request real-time re-evaluation from the *PDP* if anomalous behavior is detected. Such continuous monitoring enables immediate security responses even after initial access is granted.

The interaction between *PDP* and *PEP* forms the backbone of Zero Trust Architecture, embodying the principle that all access requests must be thoroughly validated and continuously monitored.

2.2.2. Zero Trust Use Cases

Zero Trust Architecture (*ZTA*) is being widely adopted to enhance security across a variety of industries. As cyber threats escalate globally and cloud transformation and remote work environments become more common, the adoption of Zero Trust Architectures to overcome the limitations of traditional perimeter-based security is accelerating, especially after the U.S. government issued an executive order (EO 14028) mandating federal agencies to transition to Zero Trust. These global trends are driving technology standardization discussions and intensifying competition to develop *ZTA*-based security solutions and services across various industries.

A prominent example of a successful application of *ZTA* in this context is Google's BeyondCorp [28–30]. BeyondCorp breaks away from the traditional perimeter-based security model and enhances security by thoroughly validating all requests without distinguishing between inside and outside the network. This model dynamically grants access rights based on contextual information such as user identity, device status, location, and time, enabling employees to work securely from anywhere in the world. With the introduction of BeyondCorp, Google has dramatically improved its ability to respond to internal threats and external attacks and is considered a prime example of how the Zero Trust model can work effectively in a large-enterprise environment.

Another example of Zero Trust in action is Microsoft's implementation of Azure Active Directory and its Zero Trust Security Framework [31]. Microsoft has adopted Zero Trust principles to enhance security across its cloud environment; the platform authenticates users and devices in real time and dynamically adjusts access privileges by continuously assessing risk levels.

Government agencies such as the U.S. Department of Defense (DoD) are also adopting the Zero Trust model to protect sensitive information [32]. These examples demonstrate that Zero Trust Architecture is revolutionizing the security landscape for organizations and is becoming a core component of modern security strategies.

2.3. Trust Score-Based Access Control

To effectively implement ZTA, a mechanism that goes beyond simply verifying access requests and can evaluate threats in real time and dynamically strengthen authentication is necessary. In this process, the concept of Trust Score is essential [33].

Traditional access control approaches (*RBAC*, *ABAC*, *RBA*) manage access rights of users and resources based on static policies. However, these approaches face limitations in a Zero Trust environment due to the following reasons:

- **Limitations of static policies:** In an environment where a user's environment or circumstances can change in real time, fixed access rights are insufficient to detect or respond to new threats.
- **Lack of real-time threat assessment:** Traditional access control performs authentication only at the point of access, so it cannot reflect changes in user behavior or environmental factors.
- **Complex network environments:** The adoption of cloud, mobile, and IoT devices has blurred network boundaries, making traditional access control models difficult to apply.

Trust Score is a trust algorithm-based access control approach that quantifies a user's level of trust and dynamically adjusts access privileges in real time [34]. It provides the continuous verification and context-aware capabilities that are essential in a Zero Trust environment.

- **Real-time data analysis:** The Trust Score is updated in real time based on a comprehensive assessment of user behavior, network environment, device health, past threat history, and more.
- **Dynamic access authorization:** Access permissions are dynamically adjusted based on the Trust Score, requiring additional authentication or restricting access when trust levels are low.
- **Flexible policy management:** Because Trust Score is weighted, it can be flexibly adjusted to fit your organization's environment.

The introduction of Trust Score quantifies a user's trustworthiness, requiring them to align with access control policies and achieve a Trust Score that meets a specified threshold before accessing resources [7]. This has become a key element of access control in the implementation of Zero Trust architectures.

Table 1 analyzes the differences between traditional and Trust Score-based approaches, highlighting the security, scalability, and real-time detectability of Trust Score-based access control. Trust Score is also an effective way to maintain least privileged access based on continuous validation and real-time threat assessment, and provides high flexibility to apply customized access policies to different security requirements.

Table 1. Comparative analysis of traditional access control and Trust Score-based Zero Trust.

Access Control Methods	Pros	Cons	Application Examples	Real-Time Detection	Granting Access	Flexibility	
RBAC	Granting access based on roles	Management simplicity, policy consistency	Lack of dynamic change response	Managing user groups in the enterprise	× (Not supported)	Role-based presets	Low
ABAC	Grant access based on attributes when conditions are met	Flexible, fine-grained access control	Complicated to set up and manage policies	Cloud environments, multi-system	× (Not supported)	Attribute-based conditional granting	High
RBA	Determine access with real-time risk level assessment	Better user experience, security balance	Requires risk assessment accuracy	Online services, finance	△ (limited)	Real-time granting based on risk level	Medium
TS-based ZTA	Control access with a multi-factor-based Trust Score assessment	Dynamic access, high security	Computational complexity, high computational burden	Zero Trust Architecture, high-security enterprises, government organizations	✓ (real-time)	Dynamic granting based on Trust Score	Very high

3. Trust Score Modeling and Access Control Framework in Zero Trust Architecture

3.1. Trust Score Factors and Evaluation Criteria

Trust Score is a key mechanism for implementing dynamic access control in Zero Trust Architecture (ZTA), which evaluates various security factors in real time through a weight-based formula to produce a trust score [8]. The need for Trust Score stems from the fact that traditional static access control models (*RBAC*, *ABAC*, *RBA*) have limitations in real-time threat response and continuous authentication, as discussed theoretically in Section 2. To overcome these limitations, Trust Score uses continuous data collection and real-time analytics to dynamically assess a user's trustworthiness and make decisions to allow access, require additional authentication such as Multi-Factor Authentication (*MFA*), which requires two or more verification factors, or block access. Furthermore, Trust Score enhances security resilience by enabling adaptive policy enforcement that immediately responds to changing risk scenarios, minimizing the attack surface and effectively mitigating potential breaches.

To calculate the Trust Score, detailed items and evaluation criteria were derived for the four key factors. The following describes the main aspects of each factor.

1. **User Behavior (B):** Collect and evaluate data such as users' login patterns, unusual access attempts, login time and location, etc. This factor, summarized in Table 2, is important for detecting insider threats and preventing account takeover.
2. **Network Environment (N):** Assesses the trustworthiness of the network by looking at the IP reputation of users, VPN usage, network traffic analysis, and more. The detailed items are shown in Table 3, which is essential for detecting external attacks and network-based threats.
3. **Device status (D):** Determine the security status of a device by assessing whether it is security patched, antivirus status, and whether the device is authenticated. The evaluation criteria are described in Table 4, which is especially important in BYOD environments or remote work environments.

Table 2. User Behavior (B) Factor Details.

Detail Item	Description	Detail Item Score
① Login Frequency	Establish normal login patterns, and consider deviations from those norms as security risks (login frequency per day)	20 pts: Normal (1 to 5 times) 15 pts: Slightly abnormal (6 to 10 times) 10 pts: Suspicious (11 to 15 times) 5 pts: At risk (16+) 0 pts: Very high risk level (30+)
② Failed Login Attempts	Utilize login failure counts to detect unusual login attempts	20 pts: 0 to 2 failures (Normal range) 15 pts: 3–5 failed attempts (Some mistakes) 10 pts: 6–10 failed attempts (Abnormal patterns) 5 pts: 11–50 failed attempts (Brute force likely) 0 pts: 50+ failures (Automated attack attempts)
③ Off-Hours Login	Assess security threats by detecting users accessing different time zones than they normally log in to	20 pts: Normal business hours (9:00–18:00) 15 pts: Acceptable overtime (18:00–22:00 login) 10 pts: Unusual hours (22:00–02:00 login) 5 pts: Weekend and holiday logins (High risk) 0 pts: Very high risk time logins (02:00–05:00)
④ New Location Login	Detect anomalous behavior where users sign in from places, they do not normally sign in from	20 pts: Traditional login region (same as before) 15 pts: Different region within the same country 10 pts: New country login (low risk) 5 pts: New country login (medium risk) 0 pts: Impossible move detected
⑤ Downloading large amounts of data	Evaluate for potential attacks like account takeover or data exfiltration when users download unusually large data volumes rapidly	20 pts: Normal download (500 MB or less per day) 15 pts: Minor over download (2 GB–5 GB) 10 pts: Unusual downloads (5 GB–10 GB) 5 pts: High risk (10 GB–50 GB downloads per day) 0 pts: Data breach (50 GB or more downloads)

Table 3. Network Environment (N) Factor Details.

Detail Item	Description	Detail Item Score
① IP Reputation	Evaluate your user’s IP address to see if it’s on a malicious IP list	20 pts: Highly trusted IP 15 pts: Typical IP 10 pts: IP with warnings 5 pts: Blacklisted IP 0 pts: Need immediate access blocked
② VPN Usage	Detect VPN usage to assess external IP hiding and anonymous access	20 pts: Enterprise-approved VPN 15 pts: Trusted VPN 10 pts: Known General VPN 5 pts: Use an anonymous VPN service 0 pts: Using a known malicious VPN service
③ Anomalous Network Traffic	Detect anomalies in user traffic, analyze anomalous data transfer volumes and access frequency	20 pts: Normal traffic pattern 15 pts: Minor anomalies (slightly increase) 10 pts: Increased abnormal traffic 5 pts: Lots of abnormal traffic 0 pts: Malicious traffic
④ Access Location Reliability	Evaluate access in non-traditional countries or high-risk areas	20 pts: Registered trusted location 15 pts: Previously frequented location 10 pts: New location (low risk) 5 pts: New location (medium risk) 0 pts: Accessing high-risk countries/regions
⑤ Network Stability	Evaluate the reliability of network connectivity, frequency and duration of connection session interruptions	20 pts: Very stable (no outages) 15 pts: Slightly unstable (intermittent reconnection) 10 pts: Intermittent disconnections 5 pts: Frequent disconnections 0 pts: Very unstable or suspected DDoS

Table 4. Device Status (*D*) Factor Details.

Detail Item	Description	Detail Item Score
① Security Patch Status	Assess whether you have the latest security patches	20 pts: All security patches applied 15 pts: Some critical patches missing 10 pts: Many patches missing 5 pts: Major patches not applied 0 pts: Very poorly patched
② Antivirus Status	Evaluate whether your antivirus program is active and up-to-date	20 pts: Up-to-date and active 15 pts: Active but delayed updates 10 pts: Inactive 5 pts: Long out of date 0 pts: No antivirus installed
③ Device Authentication	Assess whether a device is pre-authenticated	20 pts: Authorized corporate-managed device 15 pts: Personally authorized device 10 pts: Unenrolled but identifiable device 5 pts: Unauthorized device 0 pts: Unidentified or illegal devices
④ Device Integrity	Assess device integrity status, such as whether it is rooted or jailbroken	20 pts: Full integrity 15 pts: Minor anomalies 10 pts: Questionable integrity 5 pts: Evidence of jailbreak or rooting attempts 0 pts: Obvious integrity compromise
⑤ Device Encryption	Evaluate whether your device applies encryption when storing and transmitting data	20 pts: Fully encrypted 15 pts: Only sensitive data is encrypted 10 pts: Partial or limited encryption 5 pts: No encryption applied 0 pts: High risk of sensitive data leakage

4. **Threat History (*T*):** Analyzes a user’s history of previous security incidents, policy violations, etc., to assess their long-term trustworthiness. The detailed factors are presented in Table 5, and this is useful for proactively blocking repeat threat behavior.

Table 5. Threat History (*T*) Factor Details.

Detail Item	Description	Detail Item Score
① Threat History	Evaluate a user’s involvement in past security incidents	20 pts: No incident history 15 pts: Minor accident history (1 incident) 10 pts: Moderate accident history (2–3) 5 pts: Severe incident history (4 or more) 0 pts: Repeated and persistent incidents
② Security Policy Violation History	Evaluate a user’s history of security policy violations	20 pts: No violation history 15 pts: Minor policy violation (1 incident) 10 pts: Repeated minor violations (2–3 times) 5 pts: Major policy violations (1 or more) 0 pts: Serious violations requiring immediate action
③ Account Compromise History	Assess whether your account has been compromised or breached in the past	20 pts: No history of violations 15 pts: Suspected minor breach (unconfirmed) 10 pts: 1 account breach (low risk) 5 pts: Multiple breaches (medium risk) 0 pts: Serious breach history
④ Frequency of Policy Violation	Assess how often users violate security policies	20 pts: No policy violations 15 pts: Infrequent (once a year or less) 10 pts: Intermittent (2–3 times per year) 5 pts: Frequent (4 or more times per year) 0 pts: Very frequent (1 or more per month)
⑤ Response to Past Threats	Responding to past threats and evaluating possible improvements	20 pts: Immediate response and recurrence fully prevented 15 pts: Rapid response and partial prevention of recurrence 10 pts: Responded but did not prevent recurrence 5 pts: Delayed response and risk of recurrence 0 pts: No response and immediate action required

These factors are collected, analyzed, and weighted in real time to calculate a Trust Score, which is dynamically recalculated to trigger additional authentication or block access upon detecting anomalous behavior, providing a significant advantage over static models.

3.2. Mathematical Modeling and Computation of Trust Score

Trust Score (TS) is calculated through a weighted average formula that reflects the evaluation scores and weights of various security factors. The model quantifies factors of user behavior (B), network environment (N), device state (D), and threat history (T) to enable dynamic access control.

3.2.1. Trust Score Calculation Formula

The Trust Score is calculated by the following formula:

$$TS = \sum_{i=1}^n w_i \cdot F_i$$

- w_i : weight of factor i ($\sum_{i=1}^n w_i = 1$);
- F_i : evaluation score of factor i (range: 0 to 100);
- n : total number of evaluation factors (in this study, $n = 4$).

Specifically, this study includes the following factors:

$$TS = (w_B \cdot B) + (w_N \cdot N) + (w_D \cdot D) + (w_T \cdot T) \quad (1)$$

- w_B, w_N, w_D, w_T : weights of each factor (their sum equals 1);
- B, N, D, T : evaluation scores for *user behavior*, *network environment*, *device status*, and *historical threat records*, respectively.

3.2.2. Weight Assignment Rationale

The assignment of the weights ($B = 0.4, N = 0.3, D = 0.2, T = 0.1$) was grounded in three complementary perspectives.

First, existing research on dynamic trust evaluation emphasizes that user behavior and network context are the most critical and dynamic indicators for access control decisions in Zero Trust environments. Wang et al. [35] proposed a Zero Trust-based dynamic access control model that integrates user behavior and network context with reinforcement learning, demonstrating their significant contribution to trust computation.

Second, empirical evidence from the UNSW-NB15 dataset further supports this prioritization. Feature importance analysis indicates that behavior-related attributes (e.g., login failures, access anomalies) and network-related attributes (e.g., protocol states, source/destination bytes) contribute more substantially to detection accuracy compared with device state or historical threat attributes. Kasongo and Sun [36] confirmed that flow and content features are the most decisive in intrusion detection performance, aligning with the higher weight assignment for B and N .

Third, industry practice and security incident reports highlight the central role of human behavior and network vectors in real-world breaches. The 2023 Verizon DBIR [37] reported that over 70% of security incidents involved the human element, including misuse of credentials, social engineering, or anomalous user activities. This reinforces the decision to assign higher weights to B and N , as they capture the most dynamic and risk-sensitive factors in trust assessment.

Taken together, theoretical reasoning, empirical validation, and industry evidence justify assigning relatively higher weights to user behavior ($B = 0.4$) and network context

($N = 0.3$), while device state ($D = 0.2$) and threat history ($T = 0.1$) were given lower weights due to their limited discriminative power in real-time trust evaluation.

These example scores were derived based on the predefined evaluation criteria presented in Section 3.1 and were intended only to illustrate how the Trust Score formula operates under different conditions

3.2.3. Formula Calculation Examples

This section provides simplified examples of Trust Score (TS) computation, using the four core factors (B, N, D, T), their sub-factor scores (0–100), and the weighted formula in Section 3.2.1 (default weights: $w_B = 0.4$, $w_N = 0.3$, $w_D = 0.2$, $w_T = 0.1$). Final TS values are compared against thresholds (≥ 80 allow, 60–79 require additional authentication, < 60 block).

(1) Case 1: Low-risk scenario

A user connects from a familiar environment with normal behavior. The TS remains high, and direct access is granted.

- Situation: A normal pattern of access in a secure environment (e.g., internal network).
- Scoring:
 - $B = 90$ points (Rationale: Overall good, with minor changes in some items);
 - $N = 85$ points (Rationale: Overall good);
 - $D = 85$ points (Rationale: Managed devices, good condition);
 - $T = 80$ points (Rationale: no history of severity).
- Calculation: $TS = (0.4 \times 90) + (0.3 \times 85) + (0.2 \times 85) + (0.1 \times 80) = 86.5$ points.
- Result: 86.5 points (above 80) → Decision to allow access.

(2) Case 2: Medium-risk scenario

Access originates from an external network using a personal device, though behavior is otherwise normal. The TS falls into the intermediate range, requiring Multi-Factor Authentication.

- Situation: Behavior is good, but use of external public networks (N) and BYOD devices (D).
- Scoring:
 - $B = 80$ points (Rationale: Relatively good);
 - $N = 50$ points (Rationale: Public network use reduces trust);
 - $D = 65$ points (Rationale: Moderate security with BYOD devices);
 - $T = 80$ points (Rationale: Good historical record).
- Calculation: $TS = (0.4 \times 80) + (0.3 \times 50) + (0.2 \times 65) + (0.1 \times 80) = 68.0$ points.
- Result: 68.0 points (60–79 points) → Decide it requires additional authentication (MFA).

(3) Case 3: High-risk scenario

Suspicious behavior is combined with an untrusted network, non-compliant device, and poor history. The TS drops below 60, leading to access denial.

- Multiple risk events including abnormal times/login attempts (B), suspicious IPs (N), and vulnerable devices (D).
- Scoring:
 - $B = 35$ points (Rationale: Highly suspicious behavior);
 - $N = 25$ points (Rationale: Network environment is highly risky);
 - $D = 40$ points (Rationale: Device security is very weak);
 - $T = 30$ points (Rationale: Poor past history).
- Calculation: $TS = (0.4 \times 35) + (0.3 \times 25) + (0.2 \times 40) + (0.1 \times 30) = 32.5$ points.
- Result: 32.5 points (less than 60 points) → Decision to block access.

3.3. Access Control Process Using Trust Score in ZTA

The real-time enforcement of Trust Score is a key mechanism for implementing continuous security assessment and dynamic access control within a Zero Trust Architecture (ZTA). Unlike traditional access control approaches, Trust Score provides the ability to quickly respond to security threats through real-time data collection and dynamic weight adjustments. This chapter describes how Trust Score interacts with Policy Decision Points (PDPs) and Policy Enforcement Points (PEPs) to control access privileges, and how it integrates with existing security policies.

In a ZTA environment, Trust Score-based access control is accomplished through the following step-by-step process, as shown in Figure 2.

1. **Access request:** A user attempts to access a system or application.
2. **Request Intercept:** The PEP intercepts an access request and forwards it to the PDP.
3. **Data collection requests:** The PDP requests information from PIP-related systems to collect key element information (user behavior, network environment, device status, past threat history, etc.).
4. **Real-time information response:** Each PIP system responds to the PDP in real time with the status of each key element related to the user.
5. **Calculate Trust Score:** The PDP calculates a weighted Trust Score based on the collected data.
6. **Security decision:** The PDP makes a decision to allow access, require additional authentication, or block access based on the calculated Trust Score.
7. **Execute the decision:** The PEP executes the PDP's decision, and the PEP connects a session to grant access to the resource as requested by the Subject.
8. **Send resource information:** The resource responds to the user's request and terminates the session if there is no response for a period of time.

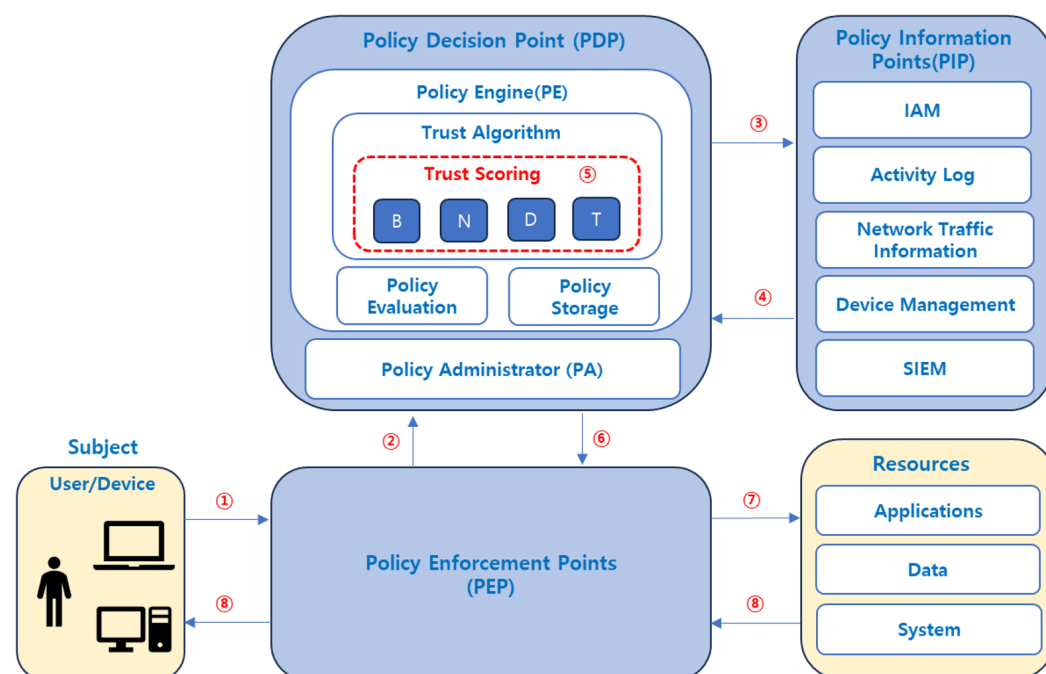


Figure 2. Trust Score access control process in a ZTA environment.

This flow is a mechanism that practically implements Zero Trust Architecture. In addition, through continuous monitoring, the *TS* is continuously re-evaluated while the session is maintained, and immediate response is taken when anomalies are detected.

Trust Score flexibly integrates with existing security policies to enhance security. Existing security mechanisms such as Multi-Factor Authentication (MFA), network isolation, and data encryption can be combined with Trust Score-based access control to dynamically adjust policies based on context. For example, you can improve the user experience by skipping additional authentication when the Trust Score is above a certain threshold, or automatically require MFA or enforce network isolation when it's below the threshold. This dynamic policy enforcement optimizes the user experience by minimizing unnecessary authentication steps while maintaining security levels.

You can also set weights and adjust thresholds to optimize policies to meet your organization's security needs. For example, you can increase the network environment (N) weight for connections from certain regions or network environments to strengthen your response to external threats.

3.4. Benefits of Trust Score-Based Access Control

Trust Score-based access control effectively supports the core principles of Zero Trust Architecture—Continuous Verification and Least Privilege Access—and offers several advantages that overcome the limitations of traditional access control.

- **Enhanced real-time threat detection and response:** Trust Score continuously assesses the health of users and devices based on real-time data. This enables immediate detection of new threats and dynamic adjustment of access permissions as they arise.
- **Enhanced dynamic authentication:** Trust Score automatically adjusts authentication strength based on the user's trust level. Legitimate users get seamless access without unnecessary additional authentication, while suspicious users are asked for additional authentication (e.g., two-factor authentication, biometrics, etc.).
- **Flexible security policy management:** Trust Score-based systems can set customized security policies by adjusting the weights to meet enterprise needs. This ensures the flexibility of security policies for different industries (finance, healthcare, cloud, etc.).
- **Improve operational efficiency:** Automated Trust Score assessment and access control minimizes administrator intervention, reducing operational costs and reducing the complexity of security management.
- **Improve user experience:** Minimize unnecessary authentication processes for legitimate users to maintain business efficiency and increase user satisfaction.

4. Trust Score Sensitivity Analysis and Policy Simulation

4.1. Simulation Environment Setup

To verify the validity and practical effectiveness of the proposed *TS* model, we constructed a simulation environment that mirrors the operational dynamics of real enterprise networks. The purpose was to evaluate how the *TS* model responds to varying conditions in user behavior, network status, device health, and historical threat patterns.

- **Virtual Environment Configuration**

A testbed was designed to simulate realistic access scenarios for users and devices. This environment includes diverse conditions—ranging from benign to potentially malicious behaviors—to analyze the model's adaptability in different Zero Trust contexts.

- **Data collection**

During each simulation run, the Trust Score values were recorded for all four evaluation factors. Real-time data, such as behavioral logs, network events, device posture, and known threat records, were generated and used to calculate the *TS* for each simulated case.

- **Weight and Threshold Configuration**

To reflect importance, default weights were assigned to the four *TS* factors as follows:

User Behavior (*B*) = 0.4, Network Environment (*N*) = 0.3, Device Status (*D*) = 0.2, Threat History (*T*) = 0.1.

These weights emphasize the role of *B* and *N* in enabling real-time risk detection and access decisions, while assigning supportive but lower influence to more static or historical indicators like *D* and *T*.

Access control thresholds were defined as follows:

- **Allow** access if $TS \geq 80$.
- **Require multi-factor authentication (MFA)** if *TS* is between 60 and 79.
- **Block** access if $TS < 60$.
- **Simulation Method:**
 1. **Setting Variables by Factor:**
 - We set the input values for *B*, *N*, *D*, and *T* to 30, 50, 70, and 90, respectively.
 - We assigned weight values for the four *TS* factors using the default configuration of 0.4 (*B*), 0.3 (*N*), 0.2 (*D*), and 0.1 (*T*).
 - To assess factor impact, we varied the value of one factor at a time while keeping the others fixed and observed the resulting *TS* changes.
 2. **Simulation Tools:**
 - We ran large-scale simulations using data analysis tools such as Python.
 - We evaluated the responsiveness and sensitivity of Trust Score based on data from various scenarios.

This simulation configuration provided a foundation for analyzing the responsiveness, sensitivity, and policy applicability of the Trust Score model in Zero Trust environments.

4.2. Sensitivity Analysis of Trust Score by Factor Value Changes

Based on the data collected through simulation, we further analyzed the change pattern of Trust Score in more detail. We compared the response of Trust Score to changes in the value of each factor to derive specific insights for optimizing security policies.

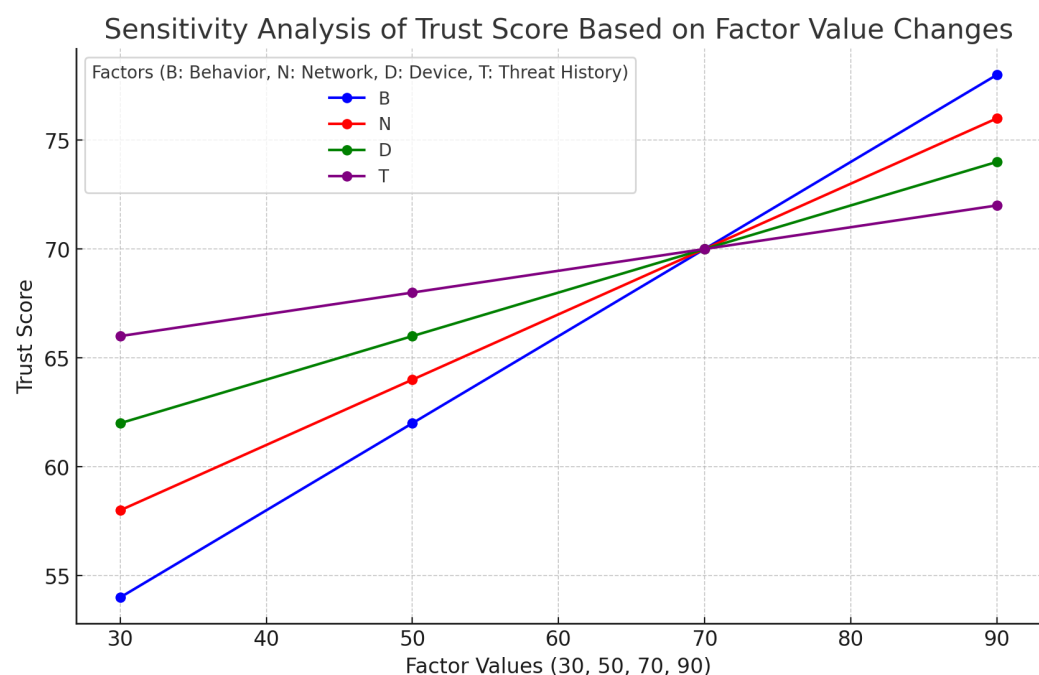
In Table 6 and Figure 3, we fixed the weights ($B = 0.4$, $N = 0.3$, $D = 0.2$, $T = 0.1$) to analyze the impact of changing the values of each factor, and numerically analyzed the impact of each factor on the Trust Score value and how sensitive it is to changing the values of each factor to 90/70/50/30.

The results of the impact and sensitivity analysis of the Trust Score value according to the change of each factor value through simulation are as follows:

- **Prioritization of User Behavior (*B*)**
 - User Behavior, set to a weight of 0.4, shows the largest change in Trust Score value (24 points) and is found to have the largest impact and the most sensitivity to changes in factor values. The higher the *B* value, the greater the increase in *TS*, which provides important insights in insider threat or account takeover scenarios.
 - Detailed analysis: A increase of 20 in the User Behavior score is associated with an average 8-point increase in *TS*. User behavior is a key factor for anomaly detection, as unusual access patterns or changes in usage behavior sensitively change the Trust Score.

Table 6. Calculated Trust Score by changing the value of each factor (weights B = 0.4, N = 0.3, D = 0.2, T = 0.1).

Baseline: 90					Baseline: 70					Baseline: 50					Baseline: 30				
B	N	D	T	T.S	B	N	D	T	T.S	B	N	D	T	T.S	B	N	D	T	T.S
90	90	90	90	90	70	70	70	70	70	50	50	50	50	50	30	30	30	30	30
<i>Varying B:</i>																			
70	90	90	90	82	90	70	70	70	78	90	50	50	50	66	90	30	30	30	54
50	90	90	90	74	50	70	70	70	62	70	50	50	50	58	70	30	30	30	46
30	90	90	90	66	30	70	70	70	54	30	50	50	50	42	50	30	30	30	38
<i>Varying N:</i>																			
90	70	90	90	84	70	90	70	70	76	50	90	50	50	62	30	90	30	30	48
90	50	90	90	78	70	50	70	70	64	50	70	50	50	56	30	70	30	30	42
90	30	90	90	72	70	30	70	70	58	50	30	50	50	44	30	50	30	30	36
<i>Varying D:</i>																			
90	90	70	90	86	70	70	90	70	74	50	50	90	50	58	30	30	90	30	42
90	90	50	90	82	70	70	50	70	66	50	50	70	50	54	30	30	70	30	38
90	90	30	90	78	70	70	30	70	62	50	50	30	50	46	30	30	50	30	34
<i>Varying T:</i>																			
90	90	90	70	88	70	70	70	90	72	50	50	50	90	54	30	30	30	90	36
90	90	90	50	86	70	70	70	50	68	50	50	50	70	52	30	30	30	70	34
90	90	90	30	84	70	70	70	30	66	50	50	50	30	48	30	30	30	50	32

**Figure 3.** Sensitivity analysis of Trust Score to changes in factor values.

- **Importance of Network Environment (N)**
 - With a weight of 0.3, the network environment has a large variation in Trust Score value (18 points) and has the second-largest impact on Trust Score after user behavior. *TS* decreases rapidly in untrusted network environments, which suggests increased security in remote work environments
 - Detailed analysis: *TS* increased by an average of 6 points when the network environment score increased by 20. This is a useful indicator for detecting external attacks or evaluating the stability of remote access environments, and shows that network environments are volatile and subject to change, which has a constant impact on Trust Score.
- **The impact of device health (D) and past threat history (T)**

- With weights of 0.2 and 0.1, respectively, device state (TS variance of 12 points) and past threat history (TS variance of 6 points) have a relatively small impact on Trust Score. However, when utilized in conjunction, these two factors contribute to balancing security policies.
- Detailed analysis: A 20-point increase in Device Health Score increased TS by 4 points, and a 20-point increase in Past Threat History Score increased TS by 2 points. Device Health Score measures device integrity and security health, and plays an important role in maintaining a trusted environment, while Past Threat History has little impact on the current Trust Score, because in a Zero Trust Architecture environment, the focus is on current health rather than past history.

This in-depth analysis confirms that user behavior (B) and network environment (N) are the key factors in Trust Score-based access control, especially in terms of enhanced monitoring for real-time insider threat detection.

4.3. Sensitivity Analysis of Trust Score by Factor Weight Changes

In this section, we quantitatively analyze how the four factors (B : Behavior, N : Network, D : Device, and T : Threat History) that make up the Trust Score (TS) in the Zero Trust policy model respond to changes in the TS when the factor values are fixed and only the weight of each factor is varied. We derive the sensitivity of each factor and quantitatively evaluate its impact on policy triggers.

The experiment consists of simulations to measure the impact of changing the weight of each factor within the Trust Score (TS) calculation model on TS . The key objective of the experimental design is to determine how sensitive TS is to adjusting the weight of each factor independently. This allows us to identify highly sensitive factors and provide quantitative suggestions on which factors to focus on in policy design.

- Setting fixed factor values:
 - B (Behavior) = 85: Login time/location/behavior falls into the normal category.
 - N (Network) = 75: Access to corporate internal network, no VPN bypass.
 - D (Device) = 65: Frequently used device, but some security warnings present.
 - T (Threat History) = 55: Some minor threat history exists.
- Conditions for changing weights:
 - For each factor (B, N, D, T), we vary the weights to 0.2, 0.4, 0.6, and 0.8.
 - For the other three factors, the remaining weights are distributed equally 1:1:1, so that the total sums to 1.

The experimental environment in Table 7 and Figure 4 was designed to determine which factor weight adjustments have the greatest impact on reaching the policy trigger boundary, based on the “average state of a real user or device”. Specifically, simulation results were calculated and plotted to quantitatively show the sensitivity of TS values to crossing policy bands (Allow ≥ 80 , MFA 60–79, Block < 60).

Table 7. TS simulation results based on flexible weight adjustment.

Weight	Behavior (B)	Network (N)	Device (D)	Threat History (T)
0.2	69	69.67	70.33	71
0.4	73	71	69	67
0.6	77	72.33	67.67	63
0.8	81	73.67	66.33	59

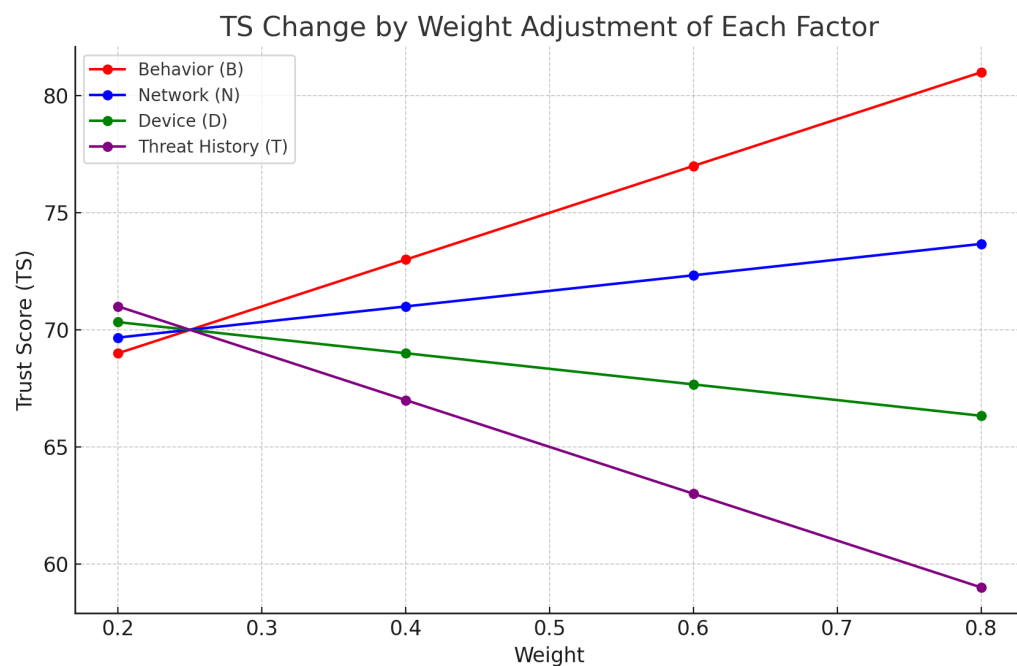


Figure 4. TS change by weight adjustment of each factor.

The results of the simulation analysis of *TS* values by changing the weight value of each factor are shown below.

In Table 8, a *TS* change of 12 or more means that changing the weight of a particular factor alone can result in a change that crosses the policy threshold (e.g., Allow ≥ 80 , Block < 60). For example, increasing the weight of *B* from 0.2 to 0.8 would increase *TS* from 69.00 to 81.00, which shows that the policy band can be shifted from *MFA* to *Allow* simply because the user has a high behavior-based trust.

Conversely, if we increase the weight of the *T* factor, *TS* plummets from 71.00 to 59.00, indicating that assigning high importance to users with a history of past threats can trigger a trigger to block access.

A *TS* change of more than 10 indicates that this factor is a highly sensitive variable for security policy decisions, and weighting is a key factor in ensuring policy agility and flexibility. Based on this sensitivity, policy designers can dynamically tune Trust Score to meet real-time security needs or service level agreements (SLAs).

Table 8. Sensitivity analysis summary.

Element	TS Variation	Sensitivity Direction	Policy Effect
Behavior (B)	+12.00	Positive	Promotes allowance for normal users
Threat History (T)	−12.00	Negative	Promotes blocking for threat users
Network (N)	+4.00	Positive	Supplements policy with environmental context
Device (D)	−4.00	Negative	Filters based on device trustworthiness

Through the simulation results, we quantitatively analyzed the sensitivity of each factor's weight change to the Trust Score (*TS*). The *B* and *T* factors have the highest sensitivity, suggesting that adjusting the weights of these factors in the policy trigger range (Allow ≥ 80 , MFA 60–79, Block < 60) has a decisive impact on the policy transition.

- The *B* factor has a sharp increase in *TS* as its weight increases, contributing to the rapid induction of a permissive policy for legitimate users.
- The *T* factor is a threat history-based factor, with a sharp decrease in *TS* as its weight increases, enabling policy to quickly block or require additional authentication.

4.4. Summary of Simulation Results

The simulations in this study served three purposes: (1) to identify sensitive elements that can act as policy triggers; (2) to verify numerical thresholds for policy decisions (≥ 80 allow, 60–79 MFA, < 60 block); (3) to suggest design directions for Trust Score-based policies suited to different industry contexts.

As summarized in Table 9, the experiments consistently showed that user behavior (B) and threat history (T) are the most sensitive factors, often shifting the Trust Score across decision thresholds, while network (N) and device (D) exert more moderate influence. This indicates that prioritizing B and T in policy design ensures timely detection of abnormal access, while N and D provide contextual balance.

Taken together, these findings highlight that the most effective policy design centers on the B factor, with flexible adjustments to N or T as needed. This approach provides a practical basis for achieving real-time responsiveness, high sensitivity, and industry-specific applicability in the TS-based Zero Trust model.

Table 9. Simulation result summary.

Perspective	Key Elements	Role	Policy Utilization
Zero Trust Operation Strategy	B, N	Real-time status evaluation and access control	User context-based policy setting, MFA inducement
Policy Design Based on Sensitivity Analysis	B, T	High TS sensitivity $\rightarrow$ Induces policy switching	Automatic trigger criteria for allow/block policies
Integrated Strategy	B (common) + N or T	Flexible weight optimization according to objectives	Adaptive policy tuning based on convenience or security enhancement

5. Empirical Evaluation of Performance and Scalability

5.1. Experimental Setup

To empirically evaluate the scalability and computational burden of the proposed Trust Score (TS) model, all experiments were conducted in a controlled desktop computing environment. The implementation and testing were carried out using a Python-based script, which is publicly available in the code repository [38], designed to process large-scale tabular data extracted from the UNSW-NB15 dataset.

The system specifications for the performance evaluation are as follows:

- CPU: Intel(R) Core(TM) Ultra 7 155H 3.80 GHz.
- Memory: 32 GB DDR4 RAM (3200 MHz).
- Storage: 1 TB NVMe SSD (PCIe 4.0).
- Operating System: Windows 11 (64-bit).
- Python Environment: Python 3.13.
- Libraries: pandas, numpy, psutil, memory_profiler, concurrent.futures.

The Trust Score computation logic was implemented using Python's data processing stack (pandas and numpy). Performance monitoring during batch processing was conducted using the psutil and memory_profiler libraries to track CPU utilization, memory usage, and processing time.

The dataset was loaded in CSV format and segmented into test batches of 10 K, 100 K, and 1 million rows. The TS values were computed per row using a weighted linear formula involving four components (B, N, D, T), derived from preprocessed features in the UNSW-NB15 dataset. In addition to UNSW-NB15, we further incorporated the CICIDS2017 dataset (Wednesday subset), which contains both benign and multiple DoS/DDoS attack traffic, to validate the generalizability of the proposed model across different environments. For CICIDS2017, a similar preprocessing pipeline was applied, and the results are compared in Section 5.2.

To evaluate the scalability, experiments were conducted with varying data sizes while measuring execution time and throughput. For assessing computational burden, CPU usage, memory consumption, and processing latency were monitored throughout the experiment.

5.2. Cross-Dataset Performance Evaluation (UNSW-NB15 vs. CICIDS2017)

To verify the generalizability of the proposed Trust Score (*TS*) model, we conducted additional experiments using not only the UNSW-NB15 dataset but also the CICIDS2017 dataset. While UNSW-NB15 includes a wide variety of network-based attack categories such as exploits, fuzzers, and denial-of-service (DoS), the CICIDS2017 dataset (Wednesday subset) contains both benign traffic and multiple DoS/DDoS scenarios that simulate large-scale flooding attacks. This combination allowed us to evaluate the model across heterogeneous network environments.

Table 10 presents the classification performance of the proposed model on the two datasets. On UNSW-NB15, the model achieved an accuracy of approximately 95.9% and an F1-score of 0.97, demonstrating strong performance under diverse attack types. On CICIDS2017, the results were nearly perfect, with accuracy, recall, and F1-score values close to 1.0. The exceptionally high performance on CICIDS2017 can be attributed to the distinct traffic patterns of DoS attacks, which make the separation between benign and malicious flows relatively straightforward.

Table 10. Performance evaluation of the proposed *TS* model on UNSW-NB15 and CICIDS2017 datasets.

Dataset	Accuracy	Precision	Recall	F1-Score	ROC-AUC
UNSW-NB15	0.9588	0.9631	0.9770	0.9700	0.9935
CICIDS2017	0.9999	0.9999	1.0000	1.0000	1.0000

These findings highlight that the *TS* model maintains consistently high performance across different datasets and attack scenarios. The results confirm the robustness and general applicability of the proposed approach beyond a single dataset. However, it should also be noted that further evaluation with IoT- and cloud-oriented datasets (e.g., Bot-IoT, TON_IoT) is required to validate the model in broader real-world contexts, which will be considered in future research directions.

5.3. Performance Analysis Under Fixed Data Volume

To evaluate the computational efficiency of the proposed *TS* model under a fixed data volume, an empirical experiment was conducted using the UNSW-NB15 dataset. The dataset includes 175,000 traffic flow records labeled with normal and attack categories. Each record was processed to calculate a *TS* value based on the model's four core elements: user behavior (*B*), network environment (*N*), device status (*D*), and threat history (*T*).

The *TS* computation was implemented in Python and executed on a standard workstation. The evaluation metrics included total processing time, per-row computation time, memory usage, and throughput measured in Trust Score computations per second (*TPS*).

Each of the four *TS* components was derived from specific features within the UNSW-NB15 dataset. Table 11 below presents the mapping between each *TS* factor—user behavior (*B*), network environment (*N*), device status (*D*), and threat history (*T*)—and the corresponding dataset features used in the calculation.

The final *TS* value is calculated using a weighted sum:

$$TS = 0.4 \times B + 0.3 \times N + 0.2 \times D + 0.1 \times T \quad (2)$$

Table 11. Mapping of Trust Score factors to UNSW-NB15 dataset features.

Factor	Meaning	Mapped Features (UNSW-NB15)	Description
<i>B</i> (User Behavior)	Behavioral patterns and connection characteristics of the user	ct_state_ttl, ct_dst_sport_ltm, is_sm_ips_ports	Reflects behavioral traits such as connection state, frequent port reuse, and multi-port/multi-IP activities
<i>N</i> (Network Environment)	Characteristics of the network protocol and connection context	proto, service, state	Represents the protocol type, service category, and session status to evaluate the trustworthiness of the network environment
<i>D</i> (Device Status)	Activity and response traits of the device	sttl, dpkts, dbytes	Indicates device behavior using TTL, the number of packets sent, and total bytes exchanged
<i>T</i> (Threat History)	Historical record of attacks or anomalies	attack_cat, label	Uses known attack categories and anomaly labels to reflect historical exposure to threats

The results in Table 12 show that the *TS* computation is extremely lightweight. The average processing time per row was only 0.0138 ms, and the total memory footprint remained under 8 MB, even when handling a dataset of substantial size. The model sustained a throughput of over 72,000 trust evaluations per second, which strongly supports its use in real-time decision-making systems within Zero Trust environments.

Table 12. Experimental result.

Data Size (Rows)	Total Time (s)	Avg Time (ms/Row)	Memory Usage (MB)	TPS (Trust Scores/s)
175,000	2.41	0.0138	7.72	72,656

5.4. Scalability Testing Under Varying Dataset

To assess the scalability of the proposed Trust Score (*TS*) model under varying dataset sizes, a series of experiments was conducted with increasing volumes: 10,000, 50,000, 100,000, and 1,000,000 records. The objective of this analysis is to validate whether the *TS* model can maintain stable performance as the number of evaluation instances grows. This scalability is a crucial requirement for real-time deployment in Zero Trust environments. As shown in Table 13 below, the results demonstrate the model's ability to sustain consistent computational performance across varying data scales.

Since the original UNSW-NB15 dataset contains approximately 175,000 rows, the 1 million record case was generated by duplicating the available dataset to simulate large-scale environments without altering its feature distribution.

Table 13. Trust Score calculation performance at different data scales.

Data Size (Rows)	Total Time (s)	Avg Time (ms/Row)	Memory Usage (MB)	TPS (Trust Scores/s)
10,000	0.14	0.0140	6.25	71,429
50,000	0.68	0.0136	6.78	73,529
100,000	1.36	0.0136	7.29	73,529
1,000,000	13.74	0.0137	9.14	72,766

The results demonstrate a linear scalability pattern:

- The average computation time per row remains nearly constant across all dataset sizes, indicating that the model's processing time increases proportionally with input volume rather than exponentially.

- Memory usage also increases gradually in proportion to the data size, showing no abrupt resource spikes even at one million records.
- Most importantly, the model achieves a throughput of over 70,000 Trust Score computations per second (TPS) in all scenarios, including the largest-scale test, which suggests that the model maintains performance even under high load.

This computational profile ensures that the *TS* model can be integrated into systems where real-time decisions are required at scale—such as user authentication gateways, micro-segmentation controllers, and security operations center (SOC) platforms—without introducing significant latency or resource overhead.

The experiments confirm that the proposed *TS* model is computationally lightweight, resource-efficient, and scalable, making it well-suited for deployment in practical Zero Trust environments that must process large volumes of contextual data continuously.

5.5. Comparative Benchmark with Legacy Access Control Models

To further validate the computational efficiency of the proposed Trust Score (*TS*)-based access control model, a comparative experiment was conducted against widely adopted legacy access control approaches. Specifically, the evaluation included Role-Based Access Control (*RBAC*), Attribute-Based Access Control (*ABAC*), and Risk-Based Access Control (*RBA*). Unlike theoretical simulations, this benchmark was performed using the actual UNSW-NB15 dataset, which contains access-level network traffic data derived from real-world scenarios. All models were evaluated on the same dataset under identical hardware conditions. Prior validation confirmed that the *TS* model maintains consistent latency regardless of data volume, ensuring the fairness and generalizability of this comparison.

Each model was operationalized with a Python-based evaluation logic:

- *RBAC*: Static role mapping and predefined permissions.
- *ABAC*: Attribute-rule matching using multiple features (e.g., protocol, flags).
- *RBA*: Dynamic risk assessment based on context and event frequency.
- ***TS (Ours)***: Weighted arithmetic computation based on four normalized features (B, N, D, T).

To ensure fairness, these implementations were executed under identical dataset size and hardware conditions. No system-level optimizations (e.g., caching or indexing) were applied, as the goal was to measure the intrinsic computational cost of each model in a uniform setting. For transparency and reproducibility, the complete implementation, including the comparative benchmarking scripts, is available in our GitHub repository [38]. The experiments were implemented using Python 3.10+, TensorFlow 2.12.0, and Scikit-learn 1.3.1.

As shown in Table 14, the proposed *TS* model significantly outperformed all legacy models in both latency and throughput. It achieved an average processing time of 0.0137 ms per record—approximately 3 to 4 times faster than the other approaches—and processed over 70,000 trust evaluations per second. These improvements are attributed to the *TS* model's lightweight computational structure, which applies basic normalization and weighted summation over a minimal number of numeric inputs. It is noteworthy that the latency result in this benchmark closely aligns with the computational burden analysis conducted earlier (0.0138 ms/record, Section 5.3), confirming the consistency of the model's performance across different evaluation scenarios. Overall, the *TS* model not only enables more adaptive and context-aware access decisions, but also imposes minimal computational overhead—making it well-suited for real-time Zero Trust deployment environments such as endpoint agents, network gateways, and security operations platforms.

Table 14. Access Control model performance comparison.

Model	Avg Latency (ms/Record)	Throughput (TPS)
RBAC	0.0490	20,416
ABAC	0.0466	21,452
RBA	0.0523	19,110
T.S. (Ours)	0.0137	70,200

Taken together, these results not only confirm the superior efficiency of the proposed Trust Score model compared to legacy access control approaches, but also demonstrate its stability across different evaluation settings. These findings provide a solid foundation for the broader discussion and future research directions presented in Section 6.

6. Conclusions and Future Research Directions

6.1. Contribution

This study makes three main contributions.

First, it defines four critical trust factors—user behavior (B), network environment (N), device status (D), and threat history (T)—and specifies 20 operational sub-metrics to compute the Trust Score. This structured framework provides the first concrete basis for aligning theoretical trust modeling with real-time decision-making requirements in Zero Trust environments.

Second, through simulation-based sensitivity analysis, we identified how each factor affects Trust Score outcomes. The results show that B and N are most sensitive for capturing dynamic user and network risks, while B and T are decisive in shifting policy triggers between allow, MFA , and block decisions. These findings offer practical guidance for assigning weights and thresholds in adaptive access policies.

Third, empirical evaluation demonstrated that the TS model is lightweight and highly scalable. It achieved low latency (0.0138 ms/record) and high throughput (>70,000 TPS), even when tested with one million records. Comparative benchmarking further confirmed its superiority over legacy models such as $RBAC$, $ABAC$, and RBA in both computational efficiency and deployment readiness. Collectively, these contributions bridge theory with operational practice and provide actionable insights for Zero Trust policy architects and implementers.

6.2. Future Research Directions

While this study presents a Trust Score-based Zero Trust policy model, further research is needed to enhance its scalability, universality, and robustness. We suggest the following directions:

(1) Dynamic Weight Optimization with AI/ML

Incorporating machine learning techniques (e.g., supervised or reinforcement learning) to adjust TS weights in real time will enable adaptive responses to evolving threats and user contexts.

(2) Federated Trust Score Learning

To ensure data privacy across organizations, applying federated learning can allow collaborative TS model training without sharing raw data, thereby supporting secure cross-domain deployments.

(3) Exploring Non-Linear Trust Score Models

While the current Trust Score model adopts a linear weighted approach for simplicity and real-time operability, it may not fully capture complex interactions among factors. For example, simultaneous risks in user behavior and network environment could escalate

nonlinearly rather than additively. Future work will explore non-linear functions (e.g., logistic, exponential) and AI-based methods (e.g., neural networks, ensemble learning) to better reflect such compounding effects while maintaining interpretability.

(4) Cross-Dataset Validation for Universality

Although this study evaluated the model with UNSW-NB15 and CICIDS2017 datasets, further validation using IoT- and cloud-oriented datasets such as Bot-IoT and TON_IoT is necessary. These additional experiments will strengthen the universality and robustness of the proposed approach, ensuring applicability to emerging network environments

(5) Integration with Physical- and Link-Layer Security

Future work will also consider integrating *TS* with physical- and link-layer security mechanisms (e.g., encryption, secure tunneling, PHY security). This cross-layer integration will allow the *TS* model to correlate access-layer trust with anomalies observed at lower layers. In doing so, it can enhance resilience against undetected physical-layer attacks.

6.3. Conclusions

This paper presented a novel Trust Score-based access control model tailored for Zero Trust Architecture. Unlike traditional approaches that rely on static rules or predefined roles, the proposed model leverages real-time contextual information and multi-dimensional trust evaluation to deliver adaptive and fine-grained access decisions. Its structural simplicity—using a weighted sum of four normalized factors—ensures practical deployability without adding excessive complexity.

Empirical validation confirmed its feasibility for real-world environments. The model consistently maintained low latency and stable throughput even at large data volumes, and it outperformed conventional access control methods (*RBAC*, *ABAC*, *RBA*) in computational efficiency. These results demonstrate that the *TS* model can serve as a viable foundation for real-time Zero Trust deployment in performance-constrained infrastructures such as endpoints and gateways.

Ultimately, this research contributes a concrete step toward operationalizing Zero Trust principles. By integrating structured factor modeling, sensitivity analysis, and empirical evaluation, the proposed framework offers organizations a scalable and efficient means to enhance security resilience while maintaining flexibility and responsiveness in access control policies.

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